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HHS CMS ASTP/ONC June 16, 2025

Re: CMS RFI 2025-08701 Health Technology Ecosystem, CMS-0042-NC, RIN 0938-AV68

Drs. Mehmet Oz and Thomas Keane:

J P Systems, Inc. (JPSys) is pleased to provide information regarding the Health Technology Ecosystem. JPSys is a certified Woman Owned Small Business founded in 1983 to serve Federal IT data management needs. We provide unique professional services in the Healthcare IT ecosystem specializing in Federal Health Informatics, Clinical Terminologies, clinical data quality improvement, and healthcare policy and strategy support. We bridge the gap between the medical and IT communities, translating requirements into designs for interoperability and managing clinical data for stakeholders.

Our responses are informed by our experience as a prime contractor with over $154 million in Federal contract awards, providing clinical data management and terminology standards expertise to the U.S. Department of Veterans Affairs, Veterans Health Administration (VHA).

We help standardize clinical data and improve clinical data quality and provide services for mapping existing VHA systems and data to HL7 FHIR®'s data elements.

We participate in the Systematized Nomenclature of Medicine Clinical Terms (SNOMED CT) and Logical Observation Identifiers Names and Codes (LOINC) Standards Development Organizations to bring new term requirements and propose harmonization efforts.

Through these efforts, we have gained expertise in TECFA, USCDI, Clinical Quality Language (CQL), and HL7 FHIR APIs.

Better data enables better patient care for better patient outcomes.

If you have any questions or wish to engage in further conversation, please reach out to Jay Lyle at Jay.Lyle@JPSys.com.

Sincerely,

Jackie Mulrooney

Founder and President

TD-3.c. Regarding digital identity implementation: What impact would mandatory use of the OpenID Connect identity protocol have?

A digital ID infrastructure could facilitate testable patient matching without the use of a centrally managed national identifier. It should be piloted.

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

We would suggest that you define the blueprint and incentivize it, as was done with HITECH ACT Meaningful Use and 21st Century Cures Act.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide directory of endpoints would enable the use of FHIR as an API rather than simply a payload data model.

A TEFCA Recognized Coordinating Entity (RCE) should publish and control it. Payers/Providers should pay a subscription fee or some other mechanism as it will greatly relieve their systems from an expensive, heavy maintenance and curation burden.

TD-6. What unique interoperability functions does TEFCA perform?

TEFCA offers centralized coordination and conformance with national, standards-based data-exchange networks.

It also offers the possibility of consistent quality assessment across the ecosystem.

TD-6.b. Are there redundant standards, protocols or channels or both that should be consolidated?

National Council for Prescription Drug Programs (NCPDP), Accredited Standards Committee X12 (X12) and other specifications have some redundancies with FHIR. Where these redundancies exist, CMS can play a role in coordinating consensus. A further study of this question may be of value.

CMS should also support current efforts to define standard maps for translating between Consolidated Clinical Document Architecture (C-CDA) and FHIR.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

USCDI is intended to support interoperability by specifying a threshold set of data elements that systems must support.

1. It is technology agnostic. It is cited as a source of requirements by both Consolidated Clinical Document Architecture (C-CDA) and US Core FHIR specifications.

2. It is intended to specify only elements for which there are existing standards assets and some level of consensus.

3. It is a living specification and evolves every year.

4. It has provided a common framework for exchange of data via US Core in FHIR.

By providing a consistent set of business requirements for technical specifications, USCDI has enhanced the ability of standards designers to prioritize their work, and it provides a baseline for aligning different standards. It is a very beneficial thing for standards designers.

The limitations of USCDI include:

1. The large number of different information models in the current environment.

2. USCDI only specifies elements that have been suggested by health IT stakeholders for a future version.

3. USCDI elements are only included for which there is sufficient existing standardization that technical specifications can implement without having to invent semantic solutions.

TD-7.a. Does it contain the full extent of data elements you need?

USCDI does not contain all the data elements for an entire EHR to be fully interoperable. USCDI only contains the elements that have been suggested by health IT stakeholders and implemented broadly. Not all desired elements may be in USCDI; this is dependent on the use case for which data interoperability is necessitated and the perceived maturity of the element. We recommend providing methods or incentives for maturing data elements that are level 2 or lower and their related standards so a larger portion of EHR data can be interoperable.

TD-7.b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

Rather than utilizing a data model, USCDI was initially based on a common clinical data set, with future versions including what health IT stakeholders have submitted for inclusion. Additionally, some of the data elements have been defined at a lower level, while others have been defined at a higher level. For example, USCDI v1 started with Substance (Medication) and Substance (Drug Class) for Allergies and Intolerance. USCDI v4 introduced Substance (Non-Medication) for Allergies and Intolerance, but “non-medication” is not at the lowest level as “non-medication” could be environmental substances or food substances; both these types of substances have very different uses in a healthcare setting. Another example of data at a varying level is Care Plan. This is a broad term that could represent an unstructured note or a data class with a structured set of data elements.

TD-7.c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

USCDI should continue to evolve as consensus is built on remaining data elements.

Many items in USCDI differ in their level of **granularity**, and this can cause confusion in efforts to construct downstream specifications. Because USCDI defines requirements, not solutions, a certain amount of this is unavoidable, but the addition of requirements should undergo harmonization with existing content in both USCDI and USCDI+, and they should be specified at the lowest feasible level. Otherwise, we will continue to see unacknowledged overlap across documented requirements and variability in the specificity available to implementers. The granularity should be studied further and recommendations made by an expert Medical Terminologist.

National Council for Prescription Drug Programs (NCPDP), Accredited Standards Committee X12 (X12) and other specifications have some redundancies with FHIR. Where these redundancies exist, CMS can play a role in coordinating consensus. CMS should also support current efforts to define standard maps for translating between C-CDA and FHIR.

TD-7.d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

Language models can be extremely useful in analyzing and classifying instance data. They should also be used to assist in the process of defining the classifications to be used in analyzing instance data. But the information models and terminologies themselves must be defined by human beings if they are to be reliably recognized and managed by human beings. This governance requirement introduces significant overhead and, at least in the near term, will impede interoperability.

TD-9.a. Regarding certification of health IT: What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Software functionality should be driven by the vendors' business and providers' requirements. Government constraint is necessary at the interface, where the profit motive incentivizes vendors to limit interoperability.

If the goal is for data to be interoperable, an API that can be utilized to share data is far more beneficial than an individual system's functionality.

TD-9.b. What would be the drawbacks?

There are no drawbacks.

TD-9.c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

Medical data exhibits a great deal of variability, which is why USCDI is a subset of Electronic Health Information (EHI). EHI has a 'very long tail,' meaning that there is a lot of data spread across a very large set of very small clinical categories.

The beauty of the structured record is that it can be communicated with confidence in predictable ways. These channels support quality monitoring, and they allow the data to be presented in consistent ways to providers and to Clinical Decision Support (CDS) routines.

The problem is that there may be relevant information not addressed in the structured data. EHI covers (almost) everything else. Since EHI is covered by information blocking regulation, it must be transmitted, but the form of transmission is undefined.

Unstructured data requires either an extensive investment of provider time or use of LLMs for processing. Investments will be necessary in both maturing AI tooling and in expanding USCDI.

However, this does not mean that stewardship of the record should be transferred wholesale to AI. Humans are responsible for the decisions, and they must retain responsibility for understanding the information.

The use of AI to analyze unstructured data brings significant challenges, primarily around quality assurance and privacy. LLMs cannot be used until they can be used responsibly. To be clear, we are not anti-AI. It is a tool, not a panacea, which must be used in a properly governed manner.

TD-9.d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

CMS should incentivize API-based data requests as CMS did with its incentives for HITECH ACT Meaningful Use and 21st Century Cures Act compliance. Provide the blueprint and monetize both an incentive and financial penalty for non-compliance. Penalties should be minimal.

TD-9.e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

Quality measures should be constructed with the data available in the EHR or EHRs modified to support the data elements necessary. The burden should not be on providers, but on EHR vendors to have capabilities to send data elements that are necessary for quality reporting.

Quality measure development efforts should be coordinated with Clinical Decision Support (CDS) and interoperability specifications, as supported by current HL7 direction around FHIR and Clinical Quality Language (CQL).

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

To support data access without special effort, CMS should promote data and terminology standards in EHR interoperability functions; support standard mappings of existing specifications (i.e., Consolidated Clinical Document Architecture (C-CDA) and FHIR).

Developer APIs need a standard way to interface with provider applications. This task can be supported with the FHIR data standard, but some key requirements for efficient implementation include:

- Agreement on security technologies (e.g., SMART on FHIR use of OAuth)
- Agreement on security scope definitions
- Agreement on scope of HIPAA requirements and transparency about where they end; ability for stakeholders to 'opt in'
- National (not funded by individual private firms) registries for searching, finding, and retrieving patient health records
- Agreement on standard algorithms for patient matching and transparency about which one is used
- Agreement on retrieval standards (volumes, domain-specific timeframes, pertinent negatives)
- Agreement on data quality expectations and processes for escalation and remediation of issues
- US Realm refinement of core FHIR resources and operations to support testably specific interfaces

Standards currently support exchange, but the information exchanged only has value if it has quality. One requirement is to reduce the burden by removing low-value information, e.g., by putting timeframes on the respective domains. For example, allergies are relevant long term, but medication administrations cease to be clinically relevant after a period of time. Another critical factor in managing cognitive load is de-duplication: if all ten facilities that have treated a patient know a particular fact, then a national query will return that fact ten times. Technology must be able to recognize and filter redundant information by accurately consolidating it.

For data that is relevant, quality means ensuring that it is correctly represented in the standard, so that similar facts can be presented consistently (both to providers and to Clinical Decision Support logic) and outliers can be appropriately flagged.

CMS should institute data quality requirements for real-world data, beyond EHR certification testing. Data should be rated as to how usable and reliable it is for clinical decision support on a quantitative scale. This has yet to be defined.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

TD-11.a. Should this capability be revised to specify standardized API requirements for EHI export?

USCDI should continue to evolve, but yes, consensus should be developed for exporting unstructured data not addressed by USCDI.

Standards currently support exchange, but the information exchanged only has value if it has quality. One requirement is to reduce the burden by removing low-value information, e.g., by putting timeframes on the respective domains. Allergies are relevant forever; medication administrations cease to be clinically relevant after a period of time. Another critical factor in managing cognitive load is de-duplication: if all ten facilities that have treated a patient know a particular fact, then a national query will return that fact ten times. Technology must be able to recognize and filter redundant information.

For data that is relevant, quality means ensuring that it is correctly represented in the standard, so that similar facts can be presented consistently (both to providers and to Clinical Decision Support logic) and outliers can be appropriately flagged.

CMS should institute data quality requirements for real-world data, beyond EHR certification testing. Data should be rated as to how usable and reliable it is for clinical decision support on a quantitative scale. This has yet to be defined.

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

Yes, however CMS must ensure data quality. High data quality is necessary for Clinical Decision Support and for coordinating care across facilities.

TD-15. Regarding bulk FHIR APIs:

TD-15.a. How would increased use of bulk FHIR improve use cases and data flow?

Increased use of bulk FHIR could support researchers and training of LLMs. Bulk FHIR is appropriate for batch jobs, and not so much transactional queries. Bulk or batched HL7 FHIR R4 messages may be sent asynchronously and processed off-hours, reducing the impact on transactional databases and processing systems. This should be the approach to consuming historical records for reporting and research purposes (i.e., not real time transactional records).

TD-15.b. What are the potential disadvantages of their use?

Bulk FHIR APIs may not allow real-time access to data as they are geared towards batch processing.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

FHIR is based on a RESTful paradigm that includes the use of URIs to identify resources. These URIs identify data sources, and they imply assumptions about their context of use: they are, ideally, navigable. When FHIR data is transferred over a multi-point network rather than via a point-to-point HTTP interaction, these URLs are no longer navigable. For document-like transfers of static payloads, this is not a major problem, but for API functionality, including search, the intervening network severs the navigability and semantics of the URL, blocking API functionality.

Network-facilitated exchange remains a great baseline for continuity of care, but additional capabilities, including Clinical Decision Support (CDS) and the ability to stage drill-down data requests, will require the full API.

If TEFCA were to provide efficient connectivity infrastructure, then providers could use the point-to-point architecture of FHIR to realize the functional value of the API. This infrastructure would include, at a high level, patient search and user authentication and authorization to information sources.

The U.S. should continue to support refinement of national implementation guides to facilitate interoperability.

TD-16.a. Do current rules encourage scalable network participation?

Yes, if appropriately implemented.

TD-16.b. What changes would improve alignment (for example, API unification, reciprocal access)?

If TEFCA were to provide efficient connectivity infrastructure, then providers could use the point-to-point architecture of FHIR to realize the functional value of the API. This infrastructure would include, at a high level, patient search and user authentication and authorization to information sources.

The U.S. should continue to support refinement of national implementation guides to facilitate interoperability.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

a) Fund the public infrastructure

The public infrastructure could include aspects such as patient registries, end point lookup and security authorizations.. This infrastructure would include, at a high level, patient search and user authentication and authorization to information sources. If TEFCA were to provide efficient connectivity infrastructure, then providers could use the point-to-point architecture of FHIR to realize the functional value of the API.

The U.S. should continue to support refinement of national implementation guides to facilitate interoperability.

b) Use incentives to encourage funded providers to use it.

CMS can financially incentivize providers and payers using the same incentive strategy which was used for HITECH ACT Meaningful Use and 21st Century Cures Act. Meaningful Use compliance. Provide the blueprint for standards-based interoperable exchange (e.g., CDC Messaging Schema using HL7 FHIR R4 rather than HL7 2.x and SOAP rather than REST). Define concrete stages and pay for production-proven success. One additional consideration is to avoid putting further burdens on small providers by requiring data from them which costs them money for which they do not receive adequate compensation.